

Introduction To Computing

What is Computer?

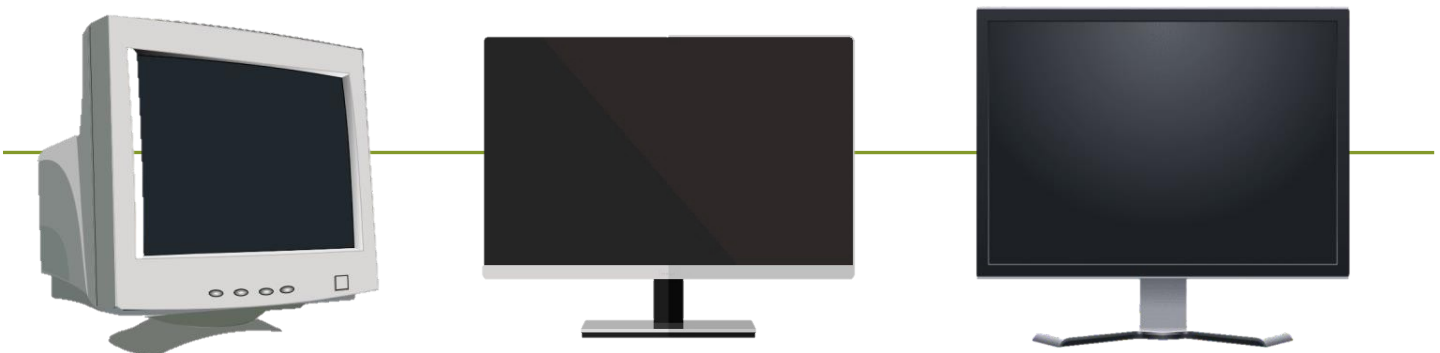
- is a **programmable** device that stores, retrieves, and processes data. The term "computer" was originally given to humans (human computers) who performed numerical calculations using calculators, such as the **abacus** and **slide rule**. The term was later given to a mechanical device as they began replacing the human computers. Today's computers are electronic devices that accept data (*input*), **process** that data, produce *output*, and store (**storage**) the results.

Desktop Computer



Monitor

- is an output device that displays information in pictorial form.



Types of Monitor

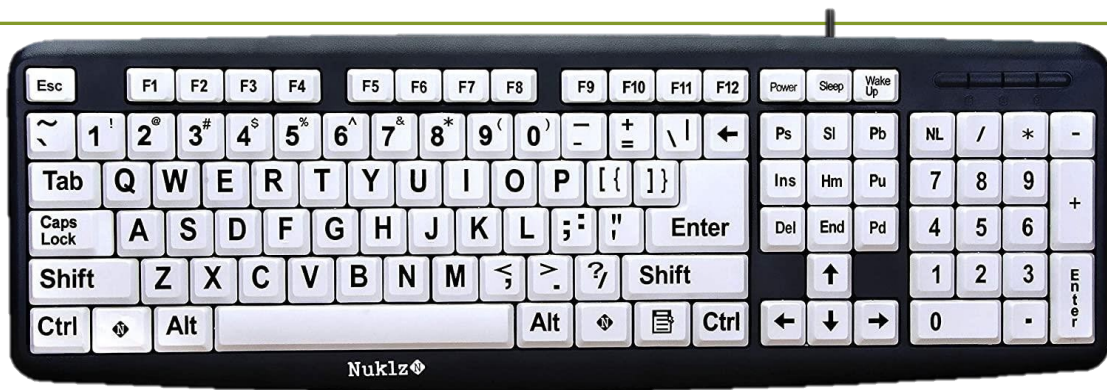
CRT – Cathode Ray Tube

LED – Light Emitting Diode

LCD – Liquid Crystal Display

Computer Keyboard

- is one of the primary input devices used with a computer. Similar to an electric typewriter, a keyboard is composed of buttons that create letters, numbers, and symbols, as well as other functions.



Computer Mouse

- is a handheld hardware input device that controls a cursor in a **GUI (Graphical User Interface)** and can move and select text, icons, files, and folders on your computer.



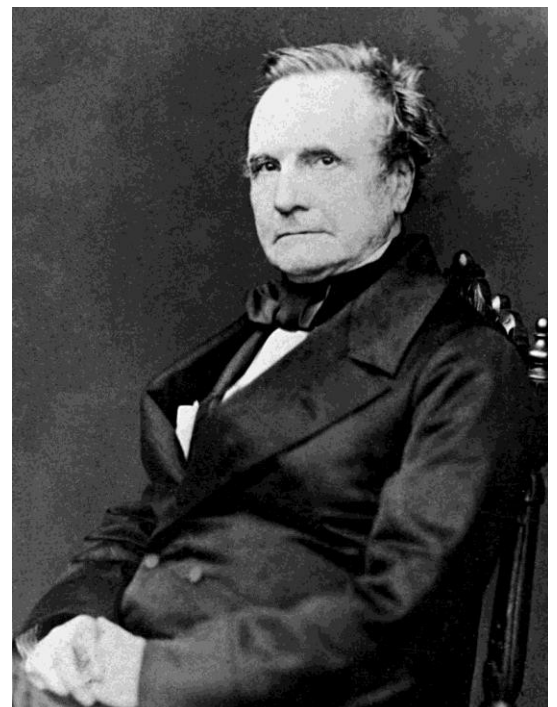
System Unit

- Alternatively referred to as a box, main unit, and system box, a system unit is the case that contains all the main parts of a computer. Below is a picture of a computer with each of the main components that help make up a computer.



Charles Babbage

- A mathematician, philosopher, inventor and mechanical engineer, Babbage originated the concept of a digital programmable computer.
- Father of the computer



Generations of Computer

First Generation – It uses **vacuum tubes**.

UNIVAC – Universal Automatic Computer

ENIAC – Electronic Numerical Integrator and Computer

EDVAC – Electronic Discrete Variable Automatic Computer

EDSAC – Electronic Delay Storage Automatic Calculator

Second Generation – It uses **transistors**.

Third Generation – It uses **Integrated circuits**.

Fourth Generation – It uses **Microprocessor**.

Fifth Generation – based on **Artificial Intelligence**.

Components of Computer

Software – The application or programs installed on a computer system.

Examples:

- **Microsoft office**
 - MS Word
 - MS Excel
 - MS Powerpoint
- **Browsers**
 - Google Chrome
 - Microsoft Edge
 - Mozilla Firefox

Hardware – The physical or tangible components of a computer system.

Examples:

- Hard Disk Drive
- Complementary Metal-Oxide Semiconductor
- CD/DVD Drive
- Motherboard
- Random Access Memory
- Etc.

Peopleware – “User”

The Objectives

- Recognize the importance or computer literacy
- Define the term. computer
- Identify the components of a computer
- Discuss the advantages and disadvantages of using computers
- Recognize the purpose of a network
- Discuss the uses of the Internet and World Wide Web
- Distinguish between system software and application software
- Describe the categories of computers
- Identify the elements of an information system Describe the various types of computer users
- Discuss various computer applications in society

World of Computers

A World of Computers

What is **computer literacy** (digital literacy)?

- Current knowledge and understanding of computers and their uses
- Computers are everywhere

Information Processing Cycle

What is the information processing cycle?

- **Input**
- **Process**
- **Output**
- **Storage**
- **Communication**

Input Device

What is an input device?

- Hardware used to enter data and instructions

Output Device

What is an output device?

Hardware that conveys information to one or more people

System Unit

What is the system unit?

- Case that contains the electronic components of the computer that are used to process data

Storage

What is storage?

- Holds data, instructions, and information for future use

Storage media

- Physical material on which data, instructions, and information are stored

Storage device

- Records and retrieves items to and from storage media

Storage

What is a USB flash drive?

- Portable storage device
- Small and lightweight enough to be transported on a keychain or in a pocket
- The average USB flash drive can hold about 2 billion characters

Storage

What is a hard disk?

- Provides greater storage capacity than a USB flash drive
- Most are housed inside the system unit

Advantages of using computers

What are the advantages of using computers?

- Speed
- Storage
- Reliability
- Consistency
- Communications

Disadvantages of using computers

What are the disadvantages of using computers?

- Violation of Privacy
- Public Safety
- Impact on Labor Force
- Health Risks
- Impact on Environment

Networks and the internet

What is a network?

Collection of computers and devices connected together, often wirelessly

- Used to share
 - Resources
 - Hardware devices
 - Software programs
 - Data
 - Information
 - Saves time and money

Networks and the internet

What is a server?

- A server controls access to resources on a network
- Clients, or workstations, request resources from the server

Networks and the internet

What is the Internet?

- Worldwide collection of networks that connects millions of businesses, government agencies, educational institutions, and individuals

Networks and the internet

Why do users access the Internet?

- Communications
- Research and Information
- Shopping
- Banking and Investing
- Classes
- Entertainment
- Download Music
- Share Information

Networks and the internet

What is the Web?

- Billions of documents, called **Web pages**, available to anyone connected to the Internet
- A Web site is a collection of related Web pages
- You can share information on a social networking Web site or a photo sharing community
- A podcast is recorded audio stored on a Web site that can be downloaded
- A Web page contains
- text, graphics, audio, video, and links to other Web pages
- A blog consists of text-stamped articles in a journal format

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Computer Software

What is software?

- Consists of a series of instructions that tells the computer what to do and how to do it
- Also called a program

Computer Software

What is a graphical user interface (GUI)?

- Allows you to interact with the software using text, graphics, and visual images such as icons
- Controls how you enter data and instructions and how the screen displays information

Computer Software

What is system software?

- Program that control or maintain the operations of the computer and its devices
 - **Operating System (OS)** is a set of programs that coordinates all activities among computer hardware devices
 - **Utility Programs** allow the user to perform maintenance-type tasks usually related to managing a computer, its devices or its programs

Computer Software

What is application software?

Programs designed to make users more productive

- Word Processing
- Spreadsheet
- Database
- Presentation Graphics

Computer Software

What is a **programmer**?

- Someone who develops software
 - **Programmer** writes the instructions to direct the computer to process data into information

Categories of Computers

What are the categories of computers?

- Personal Computers (desktop)
- Mobile Computers and Mobile Devices
- Game Consoles
- Servers
- Mainframes
- Supercomputers
- Embedded Computers

Personal Computers

What are the two most popular styles of personal computers.

- PC and compatibles use the Windows operating system
- Apple Macintosh usually uses the Macintosh operating system (Mac OS X)

Personal Computers

What is a desktop computer?

- Designed so all of the components fit entirely on or under a desk or table

Mobile Computers and Mobile Devices

What is a notebook computer?

- Portable, small enough to fit on your lap
- Also called a laptop computer
- Usually more expensive than a desktop computer with equal capabilities

Mobile Computers and Mobile Devices

What is a Tablet PC?

- Resembles a slate
- Allows you to write on the screen using a digital pen
- Especially useful for taking notes

Mobile Computers and Mobile Devices

What are mobile devices?

- Small enough to carry in a pocket
- Internet-enabled telephone is a "smart phone"

Mobile Computers and Mobile Devices

What is a handheld computer?

- Small enough to fit in one hand
- Used by mobile employees such as meter readers and delivery people

Mobile Computers and Mobile Devices

What is a personal digital assistant (PDA)?

- Provides personal organizer functions
 - Calendar
 - Appointment book
 - Address book
 - Calculator
 - Notepad

Consoles

What are game consoles?

- Mobile computing device designed for single-player or multiplayer video games

Servers

What types of servers are there?

- A **server** controls access to network resources and provides centralized storage
- **Mainframe** Very powerful, expensive computer that supports thousands of connected users
- **Supercomputer** The fastest, most powerful, most expensive computer. Used for applications requiring complex mathematical calculations

Embedded Computers

What is an embedded computer?

- A special-purpose computer that functions as a component in a larger product

Elements of an Information System

What are information system elements?

- Hardware
- Software
- Data
- People
- Procedures

Examples of Computer Usage

What software is available for a home user?

- Personal finance management
- Web access
- Communications
- Entertainment

Examples of Computer Usage

What software is available for a small office/home office (SOHO) user?

- Productivity software
- Specialty software
- Web usage
- E-mail

Computer Applications in Society

What are some examples of computer applications in society?

- Education
- Finance
- Government
- Health Care
- Science
- Publishing
- Travel
- Manufacturing

Introduction To Computing [Finals]

COMPUTER SECURITY

PROTECT

Your Computer, Your Family, and Yourself

The Internet = A World of Opportunities

Look what's at your fingertips

- A way to communicate with friends, family, colleagues
- Access to information and entertainment
- A means to learn, meet people, and explore

Online Security Versus Online Safety

- **Security:** We must secure our computers with technology in the same way that we secure the doors to our homes.
- **Safety:** We must act in ways that help protect us against the risks that come with Internet use.

Primary Online Risks and Threats

To Computers

- Viruses
- Worms
- Trojans
- Spyware

To Families

- Cyberbullies
- File-sharing
- abuses
- Invasion of privacy
- Disturbing content
- Predators

To Personal Information

- Online fraud and phishing
- Hoaxes
- Identity theft
- Spam

Primary Threats to Computer Security

Viruses/Worms

- Software programs designed to invade your computer, and copy, damage, or delete your data.

Trojans

- Viruses that pretend to be helpful programs while destroying your data, damaging your computer, and stealing your personal information.

Spyware

- Software that tracks your online activities or displays endless ads.

Primary Online Risks for Children**Cyberbullies**

- Both children and adults may use the Internet to harass or intimidate other people.

File-share Abuse

- Unauthorized sharing of music, video, and other files may be illegal, and download malicious software.

Disturbing Content

- If kids explore unsupervised, they could stumble upon images or information you may not want them exposed to.

Predators

- These people use the Internet to trick children into meeting with them in person.

Invasion of Privacy

- If kids fill out online forms, they may share information you don't want strangers to have about them or your family.

Primary Threats to Personal Online Safety**Phishing**

- E-mail sent by online criminals to trick you into going to fake Web sites and revealing personal information

Spam

- Unwanted e-mail, instant messages, and other online communication

Identity Theft

- A crime where con artists get your personal information and access your cash and/or credit

Hoaxes

- E-mail sent by online criminals to trick you into giving them money

Steps You Can Take

Your computer

- Turn on Windows Internet firewall.
- Use Microsoft Update to keep Windows up-to-date automatically.
- Install and maintain antivirus software.
- Install and maintain Microsoft Windows Defender.

Your family

- Talk with your kids about what they do online.
- Set clear rules for Internet use.
- Keep personal information private.
- Use family safety settings in Microsoft software.

Yourself

- Practice Internet behavior that lowers your risk.
- Manage your personal information carefully.
- Use anti-phishing and anti-spam technology built into Windows Vista, Windows XP SP2, Windows Live, and Microsoft Outlook.

Four Steps to Help Protect Your Computer

- Turn on Windows Internet firewall
- Use Microsoft Update to keep Windows up-to-date automatically
- Install and maintain antivirus software
- Install and maintain antispyware software

Turn on Windows Internet Firewall

An Internet firewall helps create a protective barrier between your computer and the Internet

Use Automatic Updates to Keep Software Up-to-date

- Install all updates as soon as they are available
- Automatic updates provide the best protection

Install and Maintain Antivirus Software

- Antivirus software helps to detect and remove computer viruses before they can cause damage.
- For antivirus software to be effective, you must keep it up-to-date.

Install and Maintain Antispyware Software

Use antispyware software, like Microsoft Windows Defender, so unknown software cannot track your online activity and potentially steal your information.

Other Ways to Help Protect Your Computer

- **Back up** your files regularly
- **Read** Web site privacy statements
- **Close** pop-ups using red "X"
- **Think** before you click

Back up Your Files

- Save to CD/DVD, a USB drive, or other external source

- Use a Web-based backup service

Think Before You Click

- Be cautious with e-mail attachments and links
- Only download files from Web sites you trust

Read Privacy Statements

- Understand what you are getting before you agree to download or share your personal information

Use the Red "X" to Close Pop-ups

- Always use the red "X" in the corner of a pop-up screen.
- Never click "yes," "accept," or even "cancel," because it could be a trick that installs software on your computer.

Take Steps to Help Protect *Your Family*

- **Talk** with your kids about what they do online
- **Set** clear rules for Internet use
- **Keep** personal information private
- **Use** family safety software

Talk with Your Kids about Online Risks

Talk frankly with your kids about Internet risks, including

- Online criminals
- Inappropriate content
- Invasion of privacy

Empower them by teaching them how their own behavior can reduce those risks and help to keep them safe when they are online

Pay Attention to What Your Kids Do Online

- Keep the computer in a central area
- Get to know how your kids use the Internet
- Let your kids be the teacher
- Teach kids to trust their instincts
- Encourage them to report any problems

Keep Personal Information Private

- Teach children to check with you before sharing personal information online
- Monitor your children's online activities
- Teach your children to report suspicious activity
- Help children choose appropriate screen names and e-mail addresses

Set Clear Rules for Internet Use

- Do not share files or open attachments

- Do not click links in e-mail
- Treat others the way you want to be treated
- Stand up for yourself
- Respect other people's property
- Never go alone to meet an Internet "friend" in person

Use Family Safety Software

- Available in Windows Vista', Windows Live OneCare Family Safety, Xbox 360
- Helps parents manage the content their children view, what they do, and who they communicate with online

Take Steps to Help Protect *Your Personal Information*

- **Practice** Internet behavior that lowers your risk
- **Manage** your personal information carefully
- **Use** technology to reduce nuisances, and raise the alarm when appropriate

Practice Internet Behaviors that Help Reduce Your Risk

- Delete spam, don't open it
- Be on the lookout for online scams
- Use strong passwords

Manage Personal Information Carefully

- Do not share personal information in e-mail or instant messages
- Use only secure and trusted Web sites
- Make sure you are where you think you are: Web sites can be faked
- Avoid financial transactions over wireless networks
- When in public, stay private

Use Anti-Phishing and Anti-Spam Technology

- Most e-mail providers and programs, like Windows Live Hotmail' and Microsoft Outlook', filter most spam
- Phishing Filter in Internet Explorer' 7 helps to block and warn about suspicious sites

If Your Identity is Stolen

- Report it
- Follow up in writing
- Change all passwords
- Place fraud alert on credit reports

Get a copy of your **credit report** and ensure your account is marked "fraud alert" and "victim's statement"

How Does Microsoft Help Improve Security and Safety?

- **Partnership**

By partnering with non-governmental organizations, other technology companies, and governments worldwide to help keep people safe online.

- **Technology**

By developing software products and tools that increase security and safety, helping people minimize their risk online and make informed choices.

- **Guidance**

By providing useful guidance that helps people understand online risks and what actions they can take.